CSE200: Online-1 on LaTeX (A1)

Instructions

- Reproduce the following article using LaTeX exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, tables, equations, figures,references are implemented as presented in the article with the appropriate LaTeX commands.

Mark Distribution

Component	**Marks**
Text Formatting	15
Lists	15
Equations	25
Tables	25
Figures	20
Total	**100**

CSE200: Online-1 on LaTeX

Your Name (Student ID)

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1 Introduction

Effective technical writing demands **precision** and *organization* to communicate ideas clearly. In this document, we illustrate how to combine **text**, *tables*, mathematical expressions, and figures to convey information in a structured and readable way.

Using **well-defined headings**, *organized lists*, and emphasized keywords helps readers navigate content efficiently. In addition, features such as font adjustments, inline notation (a_i, b_j), and thoughtful spacing enhance the professional appearance and consistency of the document.

Overall, a properly formatted technical report integrates clear sections, accurate equations, and visual aids to maximize both readability and presentation quality.

2 Key Concepts

2.1 Nested and Mixed Lists Example

1. System Setup
   - Install software packages
   - Configure hardware
   - ! Verify connections
2. Simulation Steps
   (a) Initialize parameters

(b) Run simulation

- Monitor CPU usage

→ Record intermediate results

(c) Analyze outputs

3. Reporting

- Prepare charts

Document observations

- Summarize findings

(a) Key insights

(b) Recommendations

3 Mathematical Expressions

Mathematical representation is crucial. Consider these examples:

$$F = k\frac{q_1 q_2}{r^2} + \frac{1}{2}mv^2 \tag{1}$$

$$\begin{aligned} g(x) &= e^x \cos(x) + \ln(1+x) \\ &= \sqrt{e^{2x} + \ln^2(1+x)} \end{aligned} \tag{2}$$

$$J = \int_0^1 x^3 e^{-x} dx$$

Piecewise function:

$$p(t) = \begin{cases} \sin(t) & \text{if } 0 \le t < \pi \\ \ln(t+1) & \text{if } t \ge \pi \end{cases}$$

$$\begin{aligned} z(x) = &\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} + \frac{1}{x+1} \\ &- \sqrt[3]{1+x^2} + \frac{\tan^{-1}(x)}{2} \end{aligned} \tag{3}$$

4 Experiment Summary Table

Sample	Measurements		Status	Comments
	Voltage (V)	Current (A)		
S1	5.2	0.8	Pass	Stable
S2	4.9	0.7	Pass	Slight fluctuation
S3	3.8	0.6	Fail	Low reading
	3.9	0.65	Re-test	Needs adjustment
S4	5.5	0.85	Pass	Good consistency

Table 1: Experimental Measurements with Multirow and Multicolumn Layout

5 Figures

5.1 Image Layout and Visualization

Visual elements enhance the clarity of technical documents by presenting information in a structured and intuitive manner technical documents.

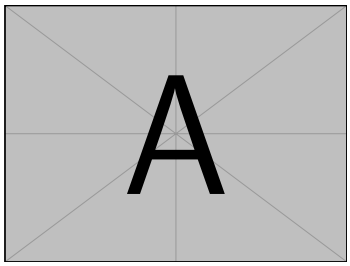
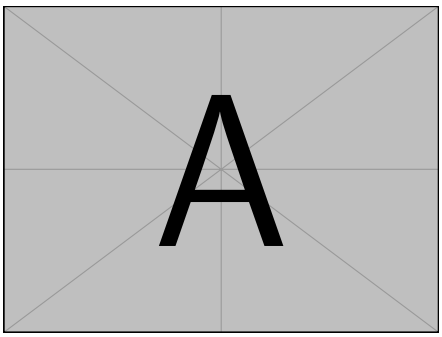


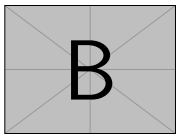
Figure 1: Wrapped image

Wrapped figures allow text to flow naturally around visual elements. This technique is useful when explanatory content is closely related to the image being displayed. Proper sizing and alignment are essential to maintain readability and avoid overlapping content. Care must be taken so that sufficient text appears alongside the figure to complete the wrapping region correctly.Here is some text describing the figure. It should be long enough to fill the height of the wrapped image. Once LaTeX "finishes" the wrap, the next sentence can be full width.

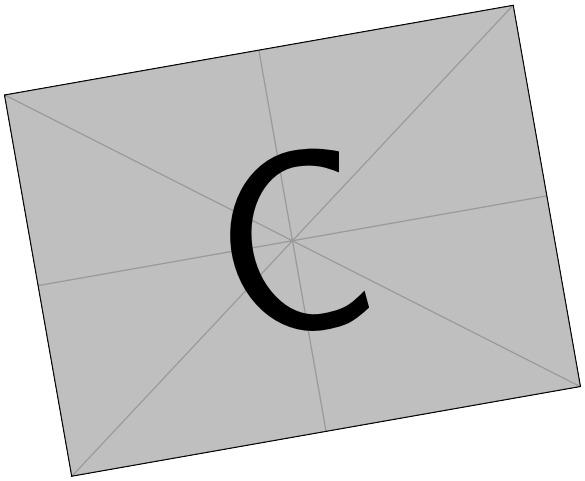
5.2 Image Layout



(a) Original image



(b) Scaled image



(c) Rotated image

Figure 2: Combined demonstration of sizing, scaling, rotation, and alignment

Figure 1 demonstrates text wrapping around an image, while Figure 2 illustrates relative sizing, scaling, rotation, multi-row layout, and subfigure alignment.

6 Conclusion

This article illustrates **effective document organization** using LaTeX with *fresh examples*, including equations, tables, and figures. Students are ex-

pected to focus on uniformity, **precise alignment**, and *proper formatting* to earn maximum marks. These additional lines provide context for explanation. They are plain text without any special formatting.